

Lee Kai Yi Zachary

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AI & Software Engineer with expertise in LLM systems, multi-agent orchestration, RAG pipelines, and full-stack application development. GPA 4.65/5.0 | SUTD Global Distinguished Scholar.

WORK EXPERIENCE

OCBC Bank

May 2026 – Aug 2026

AI Engineer Intern

Singapore

- Architected a production **wealth portfolio advisory AI agent** serving **1000+ premier private clients**, using **Retrieval-Augmented Generation (RAG)** over real-time market news and portfolio data, with a **4-agent multi-agent system** orchestrated via **LangGraph** and **AWS Bedrock** as the LLM backbone.
- Implemented **PGVector** (PostgreSQL) vector database and **ChromaDB** for local development; integrated **Redis** caching layer reducing average inference latency by **~40%**.
- Engineered high-security **AI guardrails** and **compliance controls** aligned with **MAS (Monetary Authority of Singapore)** financial AI regulations, achieving **100%** policy adherence across all model outputs.

Edgemaker.ai

Jan 2026 – May 2026

AI & Full Stack Software Engineer Intern

Singapore

- Extended an existing **multi-agent AI system** by developing **3 specialised agents**, implementing agent routing and handoff logic via **OpenAI tool calling** and **prompt engineering**.
- Integrated **OpenAI Vector Store** for document ingestion, embedding, and semantic retrieval, enabling context-aware generation over proprietary knowledge bases with **100+ documents** indexed.
- Designed and deployed scalable **RESTful APIs** using **Java (Spring Boot)** and **Node.js** with a **Next.js** frontend, serving **50+ enterprise clients** in production.

Singapore University of Technology and Design (SUTD)

May 2025 – Aug 2025

Teaching Assistant — 10.020 Data Driven World

Singapore

- Mentored 2 student cohorts in **Python** programming, data manipulation with **Pandas** and **NumPy**, and foundational **machine learning** algorithms through structured consultations and code reviews.
- Evaluated assignments and project deliverables with detailed technical feedback, driving measurable improvements in code quality and algorithmic problem-solving.

EDUCATION

Singapore University of Technology and Design (SUTD)

Sep 2023 – May 2027 (Expected)

B.Eng. Computer Science and Design, GPA 4.65/5.0, Honors with Highest Distinction

Singapore

- SUTD Global Distinguished Scholarship** Holder.
- Relevant Coursework: Data Structures & Algorithms, Computational Structures, Database Systems & Big Data, Networks and Operating Systems (OS), Elements of Software Construction, Information Systems & Android Programming.

University of Waterloo

Sep 2025 – Dec 2025

Exchange Program

Canada

- Coursework: Introduction to Machine Learning, Introduction to Artificial Intelligence, Human Computer Interactions.

PROJECTS

Chess AI Engine | Deep Reinforcement Learning, PyTorch, Policy/Value Networks, Self-Play

Jan 2026

- Built a chess AI engine using **PyTorch** and **deep reinforcement learning**, training a **neural network** via iterative self-play to evaluate board positions and predict optimal move probabilities.
- Designed end-to-end **RL training pipeline** with policy and value networks; applied experience replay to improve model convergence, stability, and generalisation across game states.

Storehouse | Node.js, Express.js, PostgreSQL, REST API, JavaScript

Dec 2025 – Jan 2026

- Engineered **RESTful** backend services with **Node.js** and **Express.js**, implementing full **CRUD** operations on a **PostgreSQL** relational database for volunteer delivery scheduling and coordination.
- Built responsive frontend features in vanilla **JavaScript**, **HTML/CSS** including an interactive Google Calendar-style delivery view and real-time form workflows.

Hotel Booking Platform | React.js, Express.js, MongoDB, Jest, Playwright, CI/CD

May 2025 – Aug 2025

- Architected a full-stack hotel booking platform with Ascenda Loyalty integration featuring advanced search with **lazy loading**, real-time sorting/filtering, and interactive map visualisation using **React.js**.
- Built secure **RESTful** backend with **Express.js** and **MongoDB** including authentication, session management, and optimised API endpoints; achieved **90%+ test coverage** with **Jest** unit tests and **Playwright** end-to-end tests.

TruthieGO | Godot Engine, GDScript, Firebase, NoSQL — DSTA BrainHack 2025 Champion

May 2025 – Jun 2025

- Engineered modular, maintainable game logic for a 2D educational board game using **Godot Engine** and **GDScript**, supporting multiple game modes and interactive mini-games; awarded **Champion** at CODE.EXP Competitive Category.
- Integrated **Firebase (NoSQL)** for real-time user authentication, session management, and cross-device progress tracking.

TECHNICAL SKILLS

AI / ML: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Systems, LangGraph, LangChain, Prompt Engineering, LLM Fine-tuning, Deep Reinforcement Learning, PyTorch, Scikit-Learn, Pandas, NumPy, Matplotlib

Cloud & Infrastructure: AWS Bedrock, Docker, Redis, PGVector, ChromaDB, PostgreSQL, MongoDB, Firebase, Hadoop, Apache Spark

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS, Tailwind CSS

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Spring Boot, Streamlit

Testing & DevOps: Jest, Playwright, PyTest, Git, Docker, CI/CD, REST API Design

Tools: VS Code, Android Studio, Figma, Godot Engine, Lucid HDL, Rhino

Methodologies: Agile / Scrum, Object-Oriented Programming (OOP), Microservices Architecture, System Design

LEADERSHIP & VOLUNTEERING

New Creation Church	Aug 2022 – Present
<i>Lead IC, Games — Kids Camp Volunteer</i>	<i>Singapore</i>
Led planning and execution of game activities for large-scale kids camps (600+ attendees); managed cross-functional volunteer teams and coordinated external vendor logistics.	
SUTD Freshman Orientation	May 2024 – Sep 2024
<i>Orientation Group Leader Head</i>	
Spearheaded event planning and people management for SUTD’s freshman orientation; mentored incoming students on academic and campus life integration.	
SUTD Volleyball Club	Sep 2023 – Jan 2025
<i>Secretary</i>	
Represented SUTD at Singapore University Games (2023 & 2024); managed club operations, attendance, and inter-club collaborations.	